Lion-Q

Q-Braid NYC-HAQ Challenge

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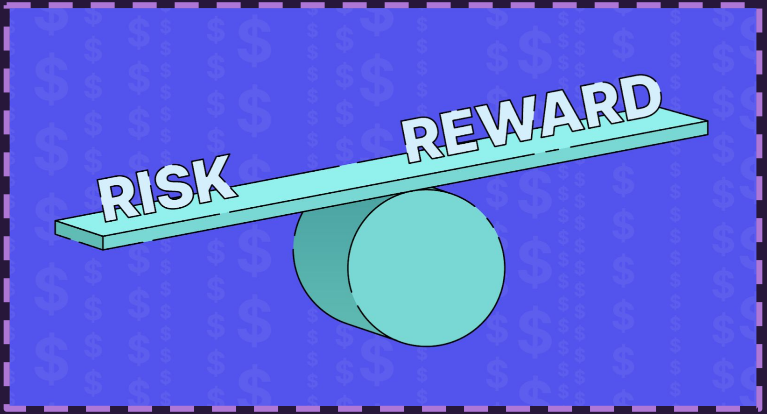
Columbia University
In the City of New York

Portfolio Optimization



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QR Code Follow Along!

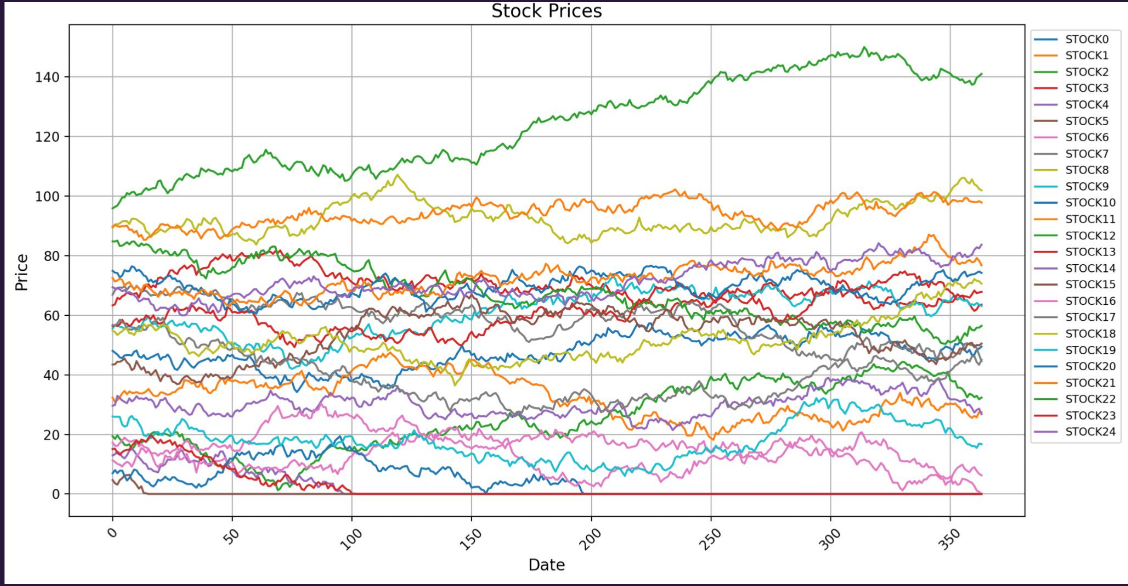
01

Methodology

Choosing a dataset and the Variational Quantum Eigensolver (VQE) Algorithm

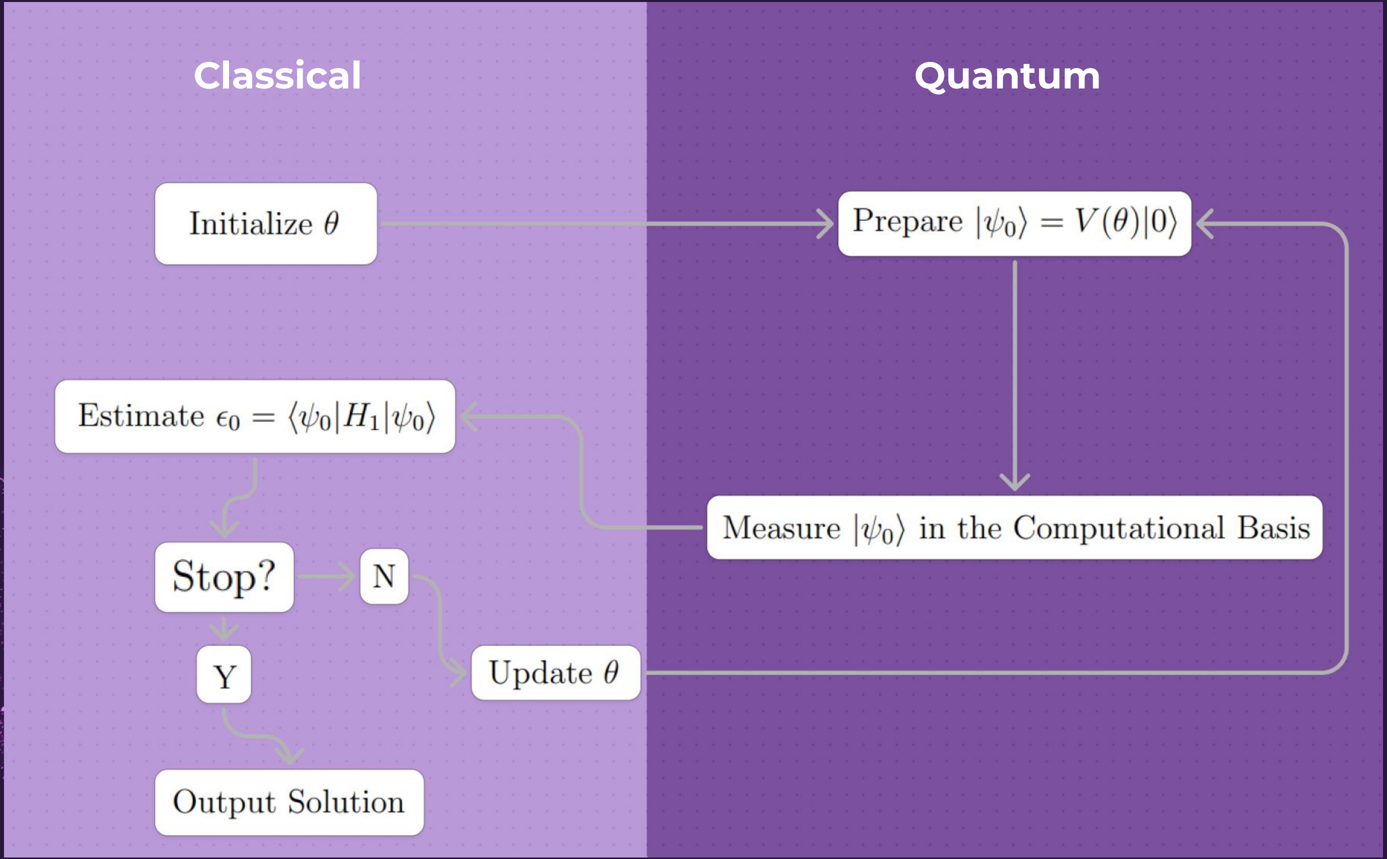
Data Collection

RandomDataProvider: Pseudo-randomly generated mock stock market data



Choose 25 stock → In our current NISQ state, simulating larger qubit systems scale exponentially

The VQE Algorithm



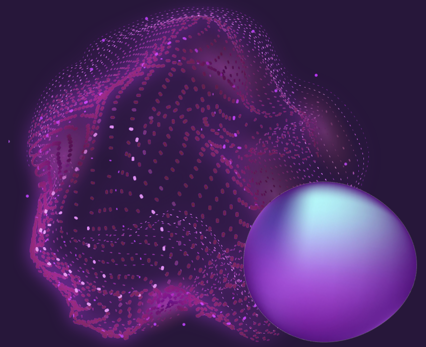
02

Designing the VQE

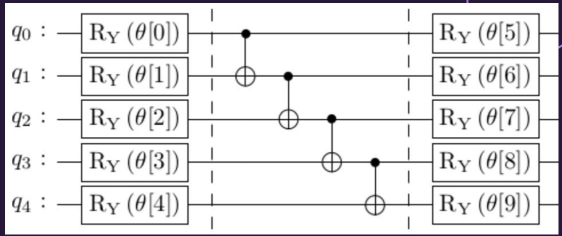
Designing the subroutines of a Variational Quantum Eigensolver

Choosing an Ansatz Circuit

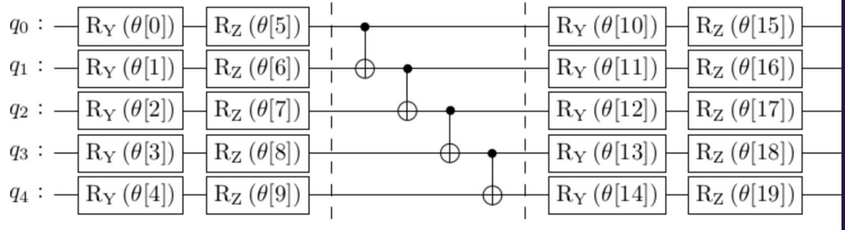
5 Qubit Examples:



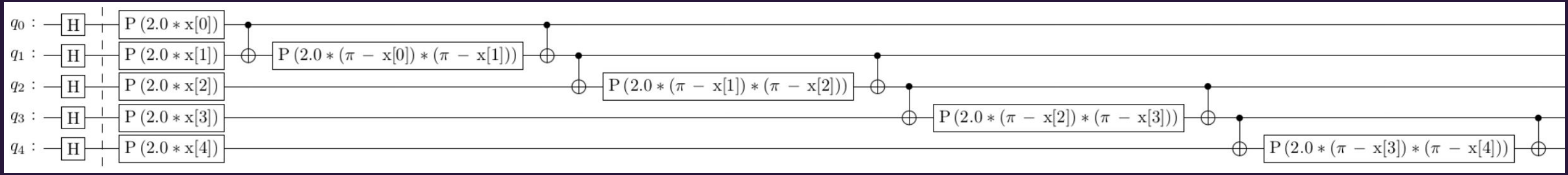
Two Local



Efficient SU(2)



ZZ Feature Map

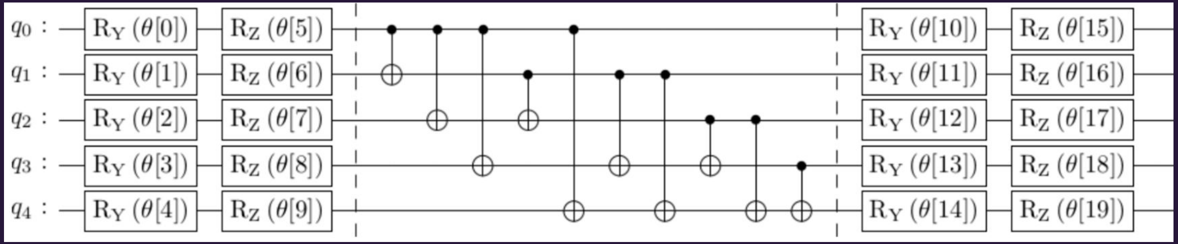


Choosing Entanglement Structure

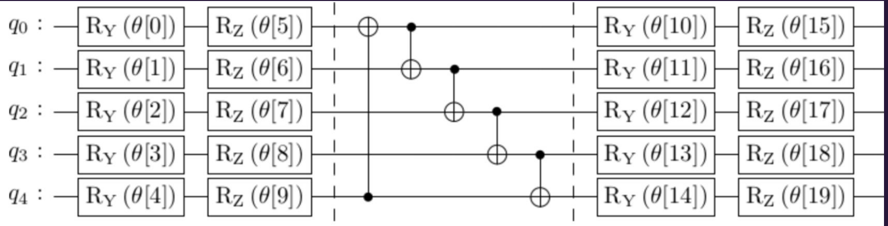
Choice: SU2 Circuit

SU2 is the fastest using this Hamiltonian!

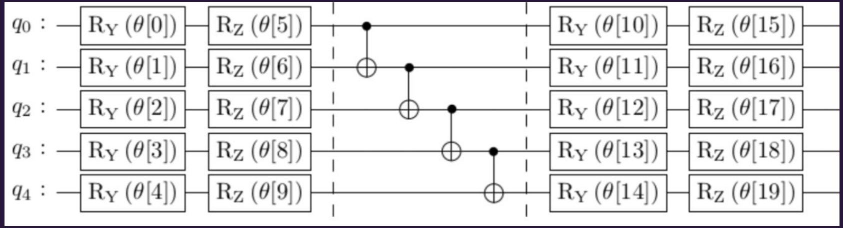
Cyclic



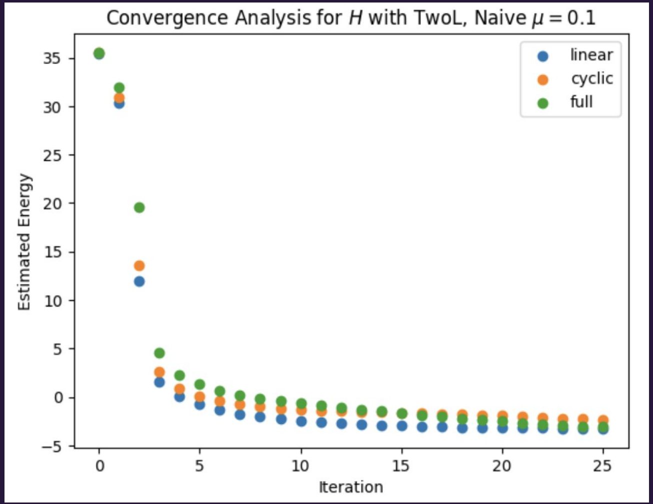
Full



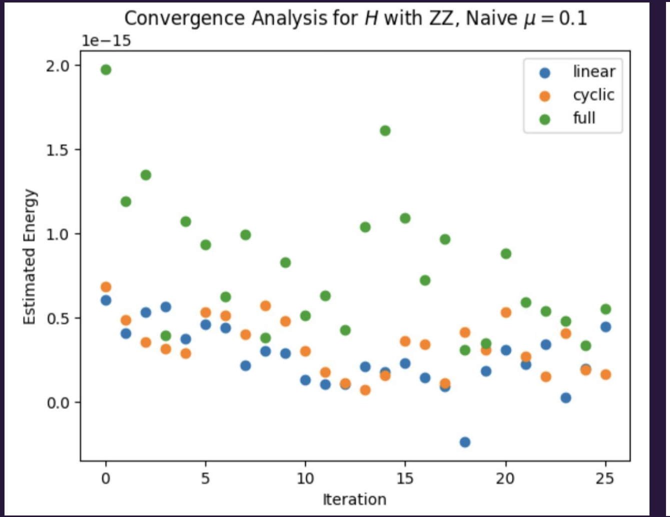
Linear

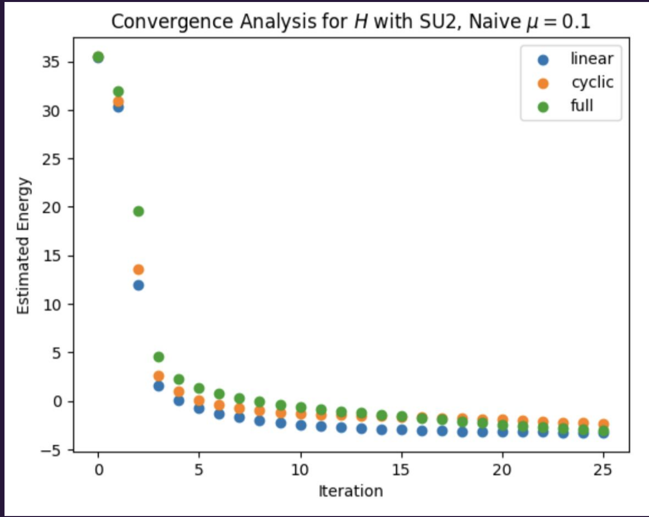


Convergence Overtime



$$H = \sum_i h_i Z_i + \sum_{i,j} J_{i,j} Z_i \otimes Z_j + \lambda \left(\sum_i \pi_i Z_i - \beta \right)^2 . \tag{16}$$



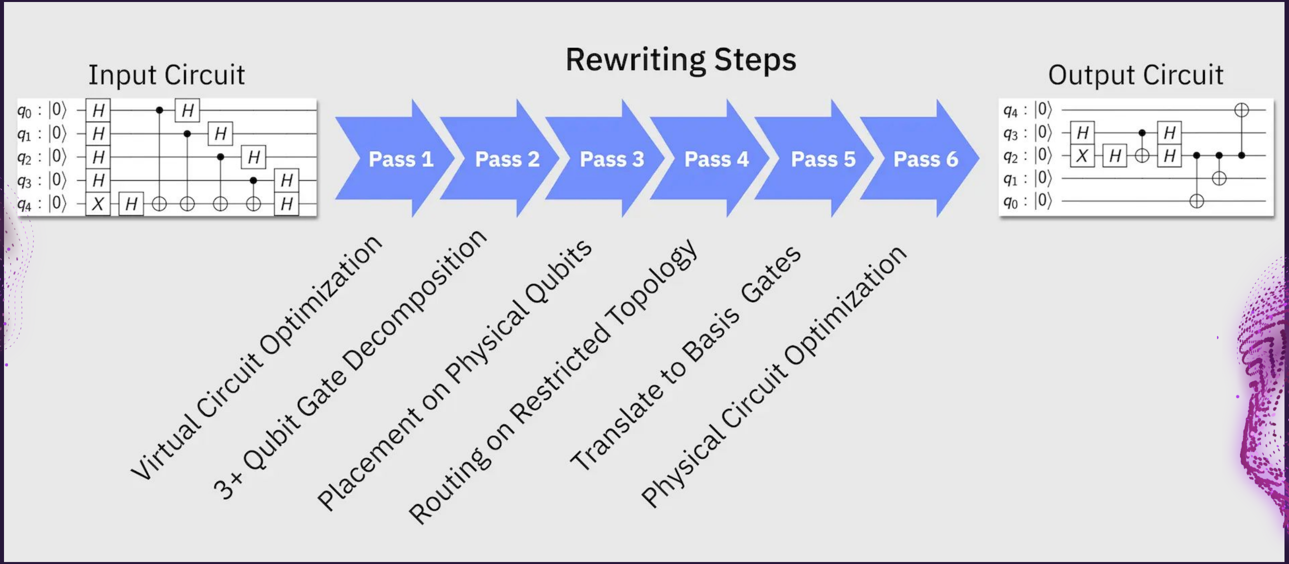


03

Quantum Transpilation

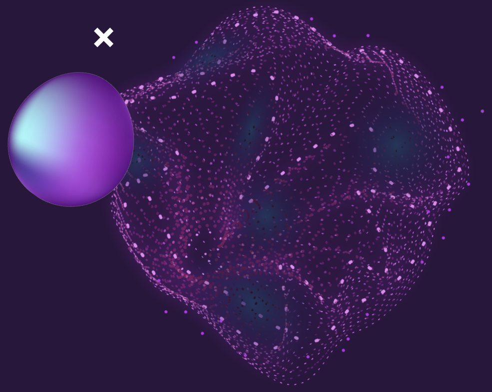
Mapping generalized quantum circuits to specific hardware

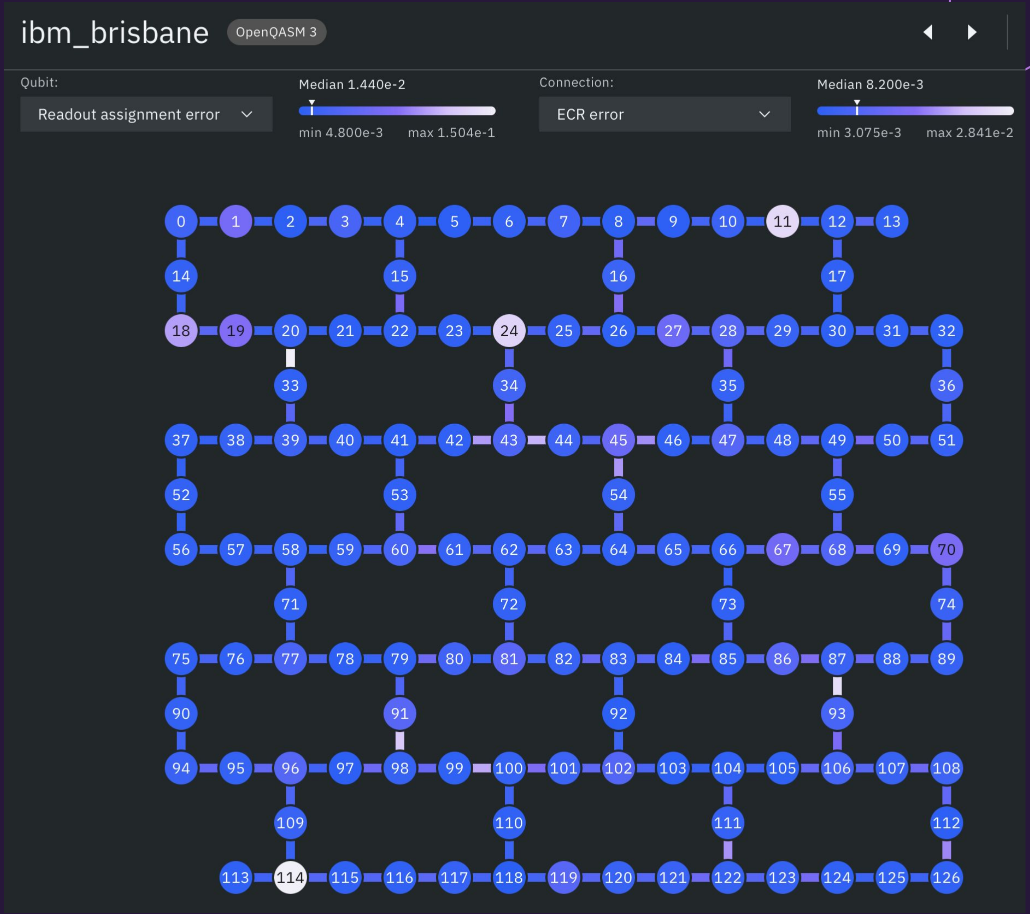
The Mapping



Transpilation is the process of rewriting a given input circuit to match the topology of a specific quantum device, and/or to optimize the circuit for execution on present day noisy quantum systems.

The Hardware



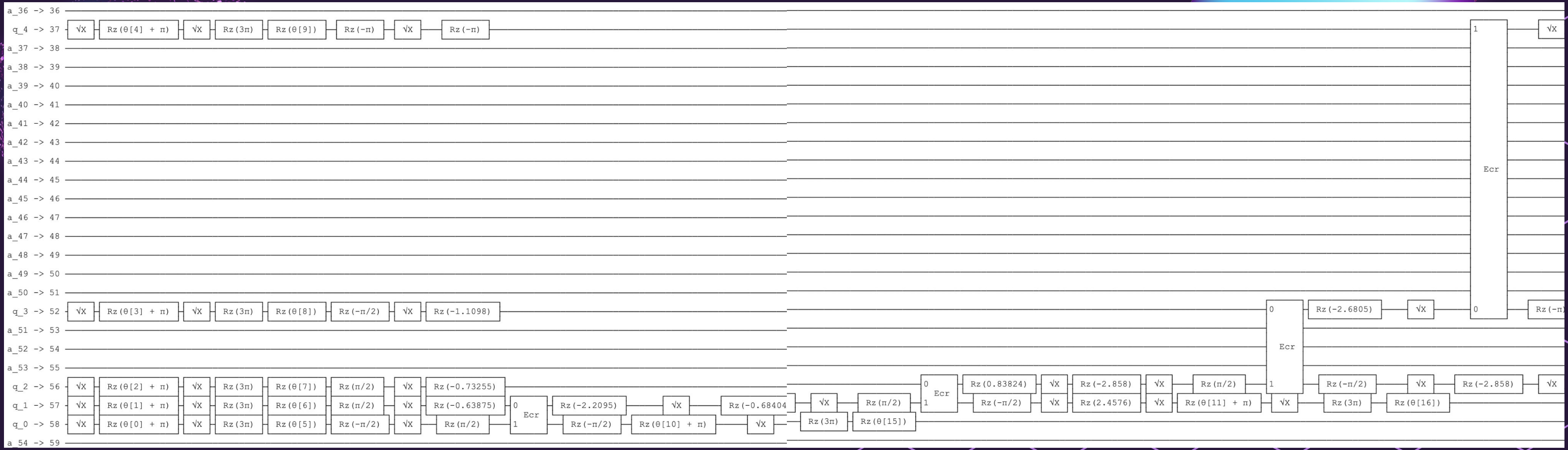


Ref: https://quantum.ibm.com/services/resources?system=ibm_brisbane

Post Transpilation

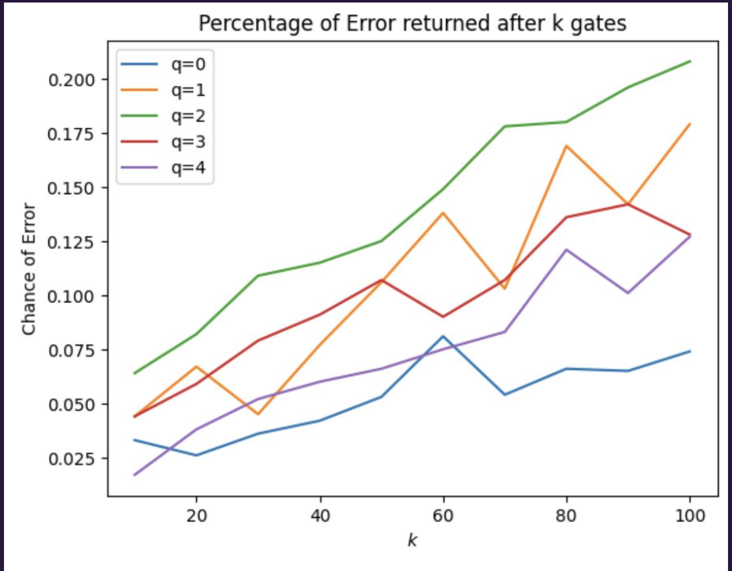
For 25 assets:
Theoretical Depth is 28 gates vs an Actual Depth of 127 gates

For the 5 qubit example from earlier:
Theoretical Depth is 8 gates vs an Actual Depth of 31 Gates for the

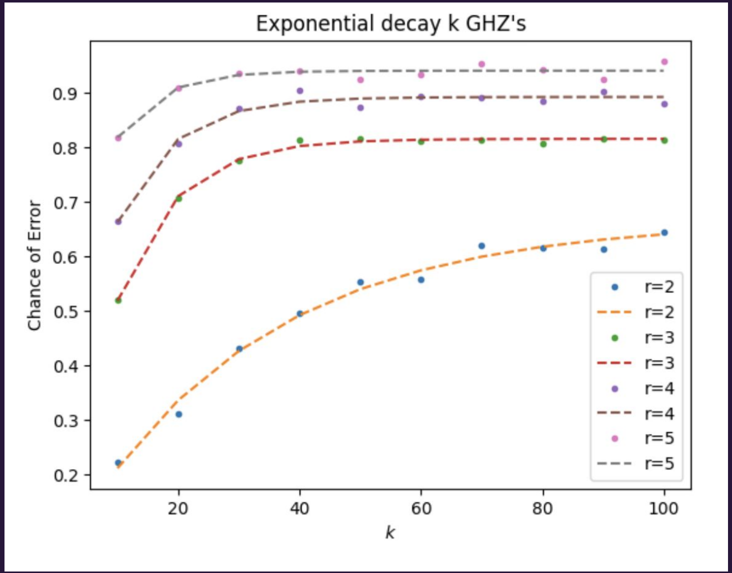


Not All Gates Are Equal

Single Qubit Gate Error Rates:
Linear Error Rate Growth



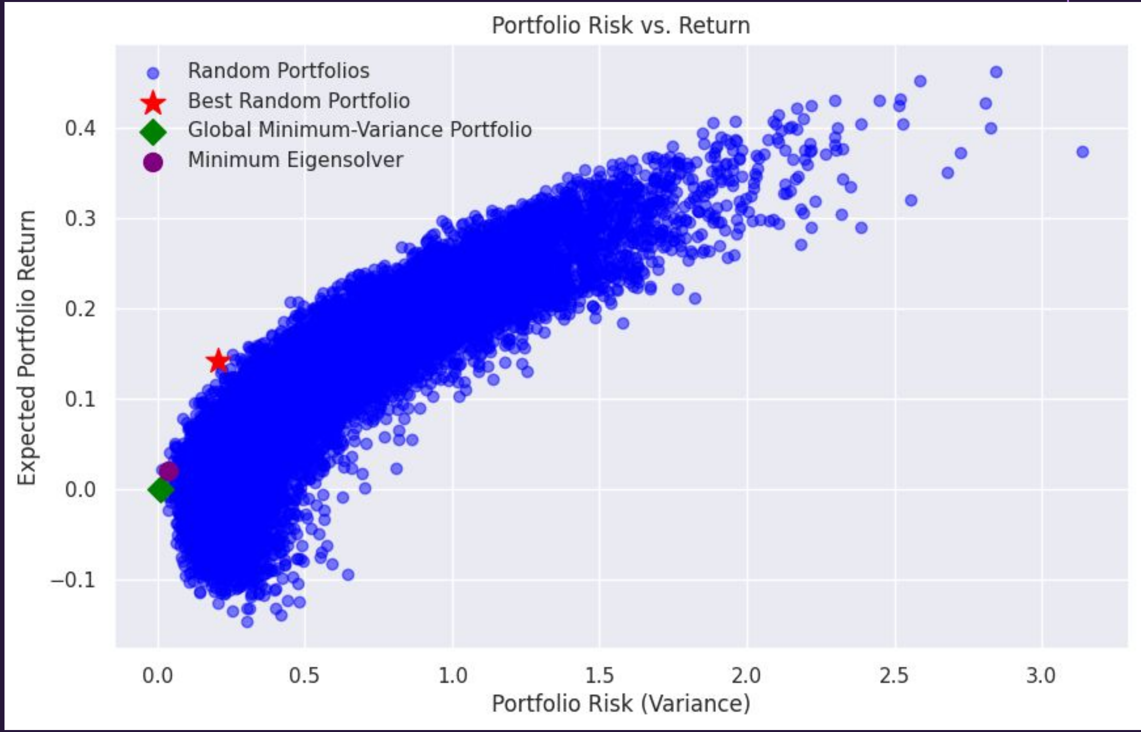
Double Qubit Gate Error Rates:
Exponential Error Rates Growth



04
Comparison with Classical Solvers

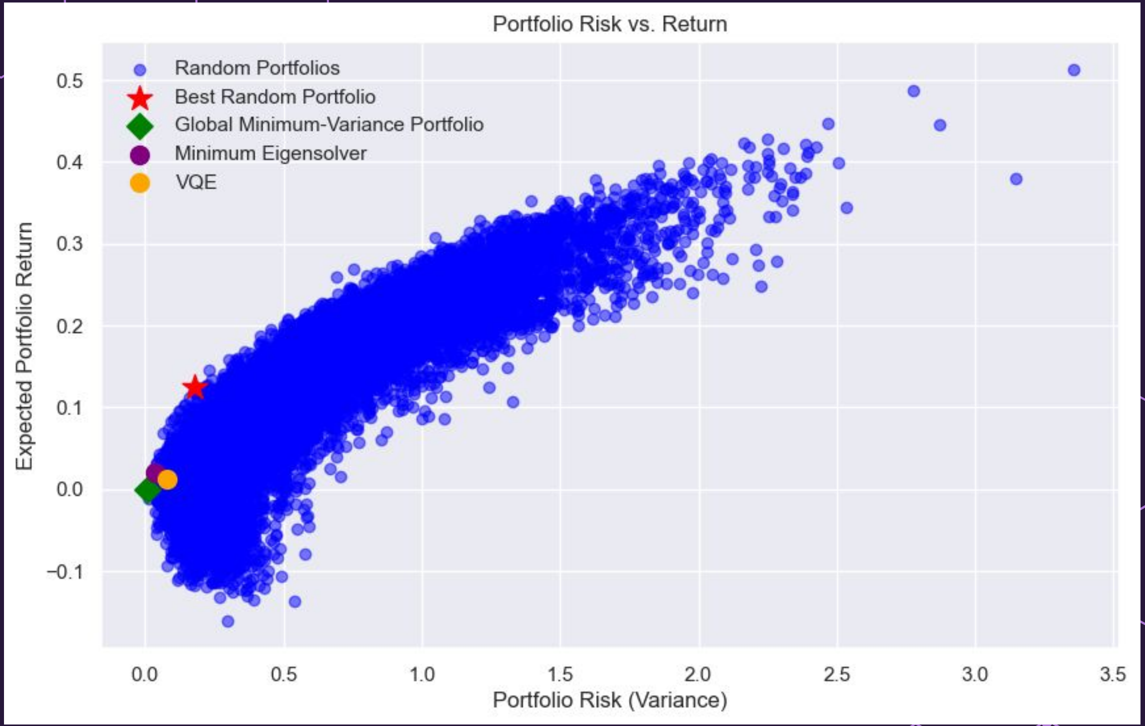
Classical Simulation

Classical Monte Carlo Simulation
Global Minimum-Variance
Minimum Eigensolver



VQE Simulation

Variational Quantum Eigensolver



05 Implications

Scalability, Optimization and Future Directions

Complexity / Scalability

Time

Classical Eigensolver: $O(n^3)$

Monte Carlo: $O(Nn^2)$

VQE: < $O(n^3)$ within BQP

$$T_{\text{total}} = \mathcal{O}\left(I \times \frac{n^2}{\epsilon^2} \times L\right)$$

Space

Classical Eigensolver: $O(n^2)$

Monte Carlo: (n^2)

VQE: $O(n)$

$$S_{\text{VQE}}^{\text{Quantum}} = \mathcal{O}(n)$$

$$S_{\text{VQE}}^{\text{Classical}} = \mathcal{O}(n^2)$$

Buonaiuto, et al. Best practices for portfolio optimization by quantum computing, experimented on real quantum devices (2023)

What Now?